

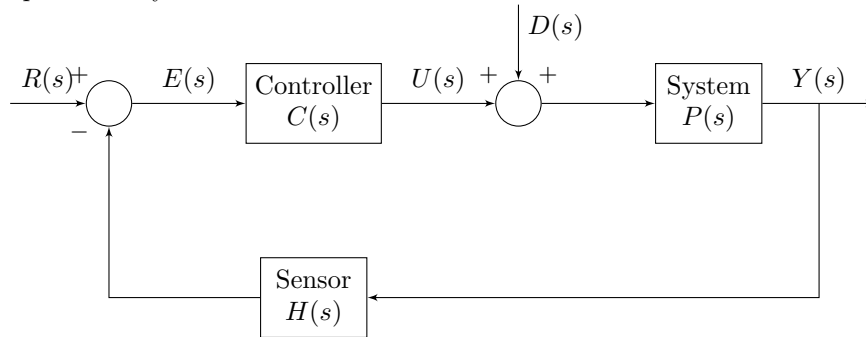
# 1 January 9, 2026 (Class 2)

System

Disturbance

Feedback loop

open close System



## 2 January 12, 2026 (Class 3)

### 2.1 Linear Constant Coefficient Ordinary Differential Equations (LCC-ODE)

A differential equation of the form

$$\sum_{k=0}^n a_k \frac{d^k y(t)}{dt^k} = \sum_{m=0}^M b_m \frac{d^m r(t)}{dt^m},$$

where  $y(t)$  is the output (response) of the system and  $r(t)$  is the input to the system; and  $a_i$  and  $b_i$  are constant.

### 2.2 Laplace Transform

The Laplace transform of a function  $f(t)$ , defined for all real numbers  $t \geq 0$ , is the function  $F(s)$ , which is a unilateral transform defined by

$$\mathcal{L}[f](s) = F(s) = \int_{0^-}^{\infty} f(t)e^{-st}dt,$$

where  $s = \sigma + i\omega$  complex frequency-domain parameter, and,  $0^-$  is the shorthand notation for

$$\lim_{\varepsilon \rightarrow 0^+} \int_{-\varepsilon}^{\infty}.$$

#### 2.2.1 Region of Convergence (RoC)